Note: When you

Q1) Circle the correct answers, and then move these ... marks):

1) Three forces are in equilibrium depending on the shown table, the magnitude of (Θ3) according (X+) is:

F	Θ	F_X	F_Y
F_1	30°	6.9282	4
F_2	130°	-4.4995	5.3623
F_3	.....	.....	.....

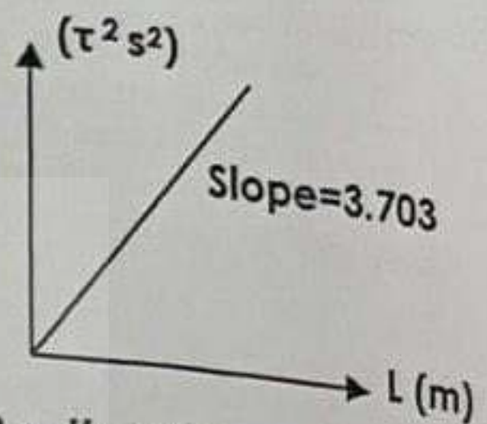
A) 84.1° B) 95.9° C) 255.5° D) 275°

2) In uniform circular motion experiment, the relation between the radius (r) of the circle and the periodic time (τ) is:

A) Inversely proportional B) Directly proportional C) no relation

3) In simple pendulum experiment, the figure below shows the relationship between the square of periodic time (τ^2) versus the length of the cord (L): The acceleration of gravity g in (m/s^2) is:

A) 10.65 B) 10.11 C) 9.62 D) 9.17



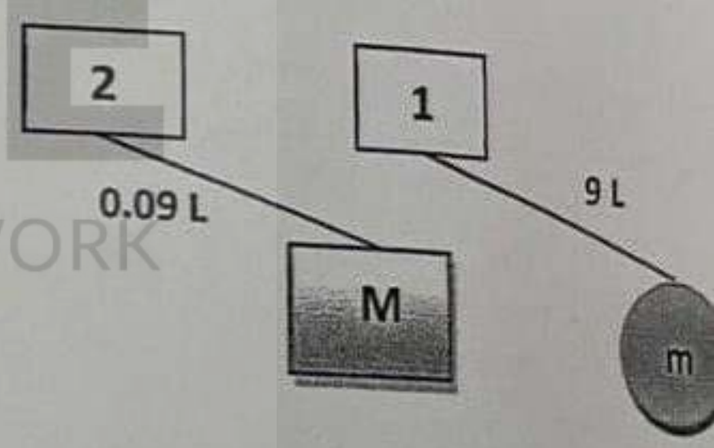
4) In uniform circular motion experiment, if the time for (20) oscillation is 9.28 s, then the angular frequency (ω) in (rad/s) is:

A) 0.116 B) 15.6 C) 14.7 D) 13.53

5) If the periodic time for pendulum (1) is (τ_m)= 0.08s) then (t_{15}) for

The pendulum (2):

A) 15s B) 0.12 s C) 18.s D) 0.18 s



6) In the simple pendulum experiment, if the length of the pendulum string increased four times, then the value of (τ) is:

A) Doubled B) tripled C) halved D) not affected

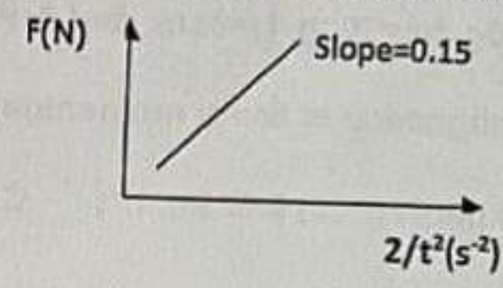
7) In a gravitational potential energy experiment, that consist of two masses (M= 1kg on the surface of th air track and the hanging mass m = 0.9 kg) slide from rest, if the final kinetic energy of the system (7.23 J). What is the initial height of the mass (m)?

A) 71cm B) 80 cm C) 13 cm D) 45 cm

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8) In Newton's second law experiment, if (M_{Total}=200g), the distance Which (M_w) moved in all trials, is:

A) 70 cm B) 80 cm C) 75 cm D) 50 cm



9) In a gravitational potential energy experiment: when you fix (m) and change the initial height (h_i), the relation between (h_i) and the time (t) is:

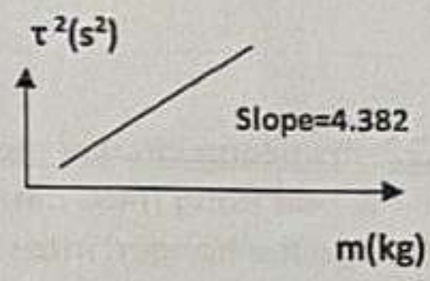
A) Inversely proportional B) Directly proportional C) no relation

10) In Newton's second law experiment, when the force is constant the relation between the hanged mass (m_h) and the traveled distance (S) by (Mw) is:

A) Inversely proportional B) Directly proportional C) no relation

11) In Hooks law experiment, the spring constant (K) is:

A) 6 N/m B) 8 N/m C) 9 N/m D) 7 N/m



12) In uniform circular motion experiment, a light string can support a stationary hanging load of mass 12 kg for keeping an object of mass m = 3.00 kg attached to the same string to rotate horizontally on uniform circular motion in a circle of radius r = 0.800 m, what is the speeds of the object?

A) 5.6m/s B) 64 m/s C) 16 m/s D) 8 m/s

13) An electric balance, when turned on, gives 0.00 grams. The uncertainty (error) in mass by this electric balance

A) 0.01 g B) 0.001g C) 0.05g D) 0.005g

14) In the physics laboratory, a student had measured the width of wood sample, and he repeated the measurement five times, if the value of [$\sum_1^5(\bar{x}-x)^2 = 0.0351$] then the standard deviation of the mean of his measurements is:

A) 0.0524 B) 0.117 C) 0.234 D) 0.094

15) ($Kg.m^2/s^2$) is unit of:

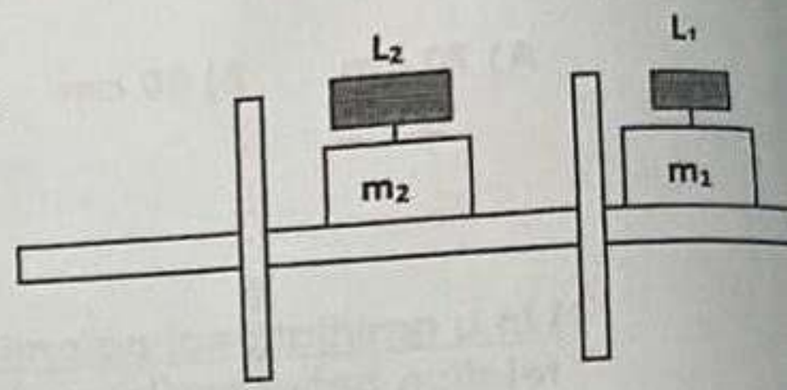
A) force B) momentum C) energy D) spring constant

16) Using significant figures rules, find (4. x 6.45):

A) 26. B) 25.8 C) 26.0 D) $3x10^1$

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* <u>In collisions experiment</u>, according to the figure Shown, answer (Q17), (Q18), (Q19):
For perfectly inelastic collision, (Where A and B are photogates) m_1 collides m_2 which in static state.
If m_1=200g, m_2=100g, L_1=3cm, and A_1=0.04s.



17) The magnitude of linear momentum for m_1 and m_2 after collision (P_f) is:

A) 0. 15 kg.m/s B) 0.02 kg.m/s C) 0.03 kg.m/s D) 0.04 kg.m/s

18) In the previous collision, we can say:

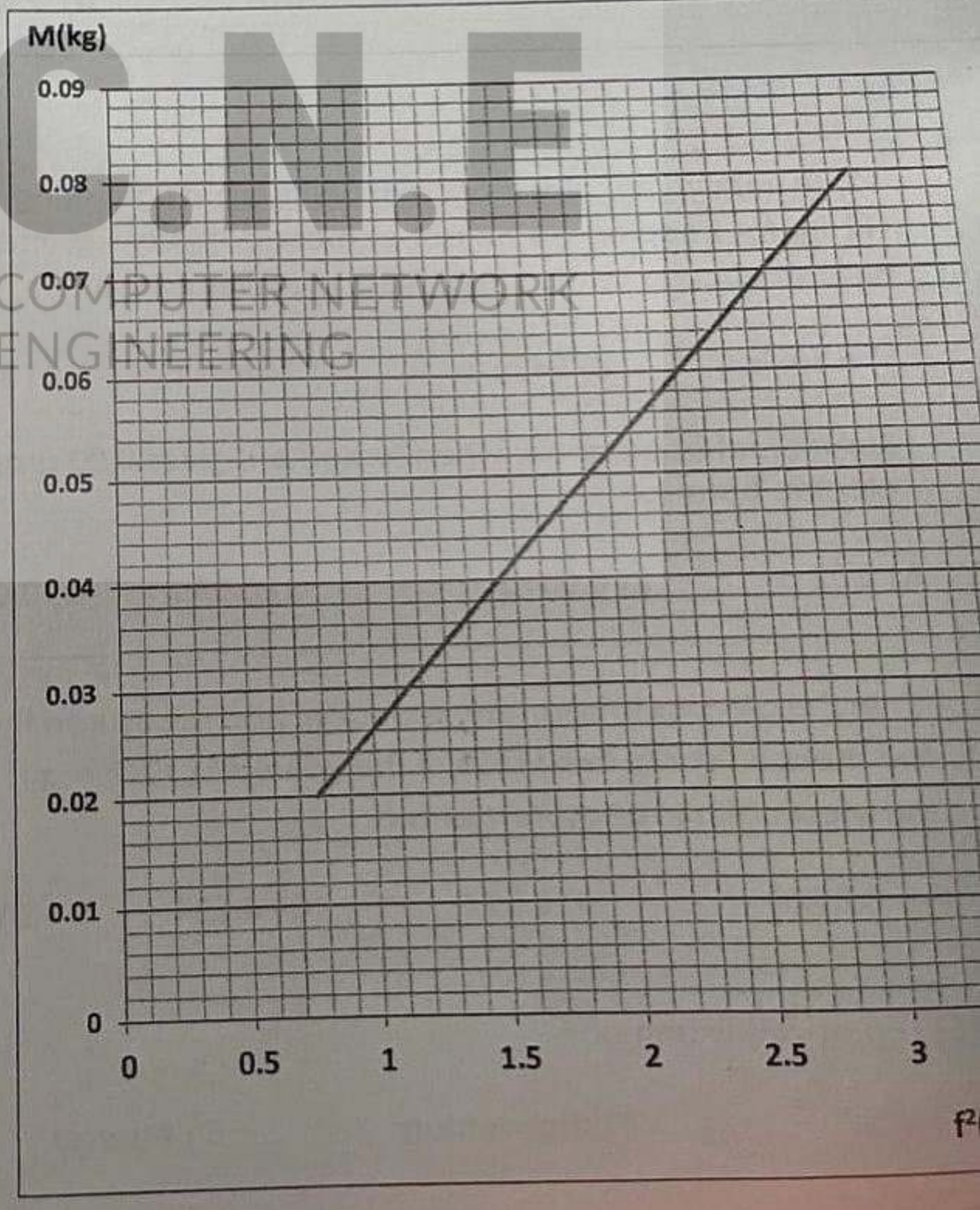
A) $V_{1f} = V_{2f}$ B) $K_i \neq K_f$ C) $P_i = P_f$ D) all of them

19) If (B_2=0.06 s),(B_1 =0.15 s). The length of the flag (L_2), which positioned on (m_2) is:

A) 6 cm B) 7.5 cm C) 4 cm D) 5 cm

Q2: *In uniform circular motion experiment:*
in the part fixing (r=50 cm), (m=15g), and change the hanged mass (M), we got the graph shown. Answer the following blanks, and then move your answers to the table:
(12 marks)

1) The physical meaning of slope
(Slope= ----------).
2) The magnitude of the slope
(Slope= ----------).
3) The magnitude of acceleration of gravity
(g= ----------).
4) At (T =0.60858s), the magnitude of, the tension in rope (T= ----------).



*** *Good luck****

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